Second Quantization

We have seen that identical bosons and fermions require symmetrized or anti-symmetrized wavefunctions under particle exchange. For systems with many particles, we already got a hint that this can get quite cumbersome (e.g., the many permutations of Fermions in the fully antisymmetric wavefunction for three particles). Things look much better in the formalism of second quantization. To get started, we will revi ew ladder operators from the harmonic oscillator, introduce the Fock space, and then discuss how we can use similar creation and annihilation operators to describe many-particle systems.

Ladder operators: you have encountered these operators before in the harmonic oscillation. Recall that the oscillator with frequency ω can be described by

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{m\omega^2\hat{x}^2}{2} = \omega\left(a^\dagger a + \frac{1}{2}\right), \tag{1}$$

where $a^\dagger$ and a are the ladder operators that raise and lower energy levels, respectively:

$$a\left|n\right\rangle = \sqrt{n}\left|n-1\right\rangle, \quad a^\dagger\left|n\right\rangle = \sqrt{n+1}\left|n+1\right\rangle, \tag{2}$$

where $\left|n\right\rangle$ is the energy eigenstate with energy $E_n = \omega(n+1/2)$. Note that $a\left|0\right\rangle = 0$, which means that the vacuum state $\left|0\right\rangle$ is the lowest energy state admissible in this system. We also usually define $N = a^\dagger a$ as the number operator, such that

$$N\left|n\right\rangle = a^\dagger a\left|n\right\rangle = n\left|n\right\rangle, \tag{3}$$

with eigenvalues $n = 0, 1, 2, \ldots$. We say that N counts the number of quanta of energy ω in the system. The Hamiltonian becomes quite transparent N: $\hat{H} = \omega(N+1/2)$.

In the oscillator, the ladder operators are defined in terms of the position and momentum operators

$$a = \sqrt{\frac{m\omega}{2}}\left(\hat{x} + \frac{i\hat{p}}{m\omega}\right), \quad a^\dagger = \sqrt{\frac{m\omega}{2}}\left(\hat{x} - \frac{i\hat{p}}{m\omega}\right), \tag{4}$$

are look more like a trick to solve the oscillator than anything else. However, they give us a hint of how to approach many-particle systems: we can think of the energy levels of the oscillator as being occupied by a certain number of quanta of energy ω, and the ladder operators as creating and destroying these quanta (the ground state is a special case). As we will see, the commutation relation of the oscillator operators $[a, a^\dagger] = 1$ is the same as the commutation relation for bosonic creation and annihilation operators (fermionic ones will anticommute instead).

Using the equation $a\left|0\right\rangle = 0$ and the definition of a in Eq. (4), derive the ground state wavefunction of the 1-D Harmonic oscillator: $\psi_0(x) = \left\langle x|0\right\rangle = (m\omega/\pi)^{1/4}e^{-m\omega x^2/2}$. Integrate your first-order differential equation $y' = xy$ with $\frac{dy}{y} = x\,dx$.

One interesting exercise is to consider now N non-interacting harmonic oscillators, which can be described by the Hamiltonian

$$\hat{H} = \sum_{i=1}^{N}\left(\frac{\hat{p}_i^2}{2m} + \frac{m\omega^2\hat{x}_i^2}{2}\right) = \omega\sum_{i=1}^{N}\left(a_i^\dagger a_i + \frac{1}{2}\right), \tag{5}$$

where a_i and $a_i^\dagger$ are the ladder operators for the i-th oscillator, satisfying $[a_i, a_j^\dagger] = \delta_{ij}$ and

$[a_i, a_j] = [a_i^\dagger, a_j^\dagger] = 0$ for $i \neq j$. The Hilbert space that encompasses these N oscillators is the tensor product of the individual Hilbert spaces, and the energy eigenstates can be written as

$$\mathcal{H} = \bigotimes_{i=1}^{N} \mathcal{H}_i, \qquad |n_1, n_2, \dots, n_N\rangle = |n_1\rangle \otimes |n_2\rangle \otimes \dots \otimes |n_N\rangle \,. \tag{6}$$

The energy eigenvalues are given by

$$E_{n_1, n_2, \dots, n_N} = \omega \sum_{i=1}^{N} \left(n_i + \frac{1}{2} \right) . \tag{7}$$

Now, if we think of the energy levels as being occupied by quanta of energy ω, we can interpret the state $|n_1, n_2, \dots, n_N\rangle$ as a state with n_i quanta of energy in the i-th oscillator.

Toy Model. This N-oscillator system turns out to be a pretty good toy model for a N-identical particle theory. To see that, let us recall what we mean by the state $|n_1, n_2, \dots, n_N\rangle$. In our usual language (which we call first quantization), we would say that the i-th oscillator is occupied by n_i quanta of energy ω, and the total energy is $E = \omega \sum_i n_i$. What **second quantization** proposes is to interpret each oscillator as a single-particle mode, and each quantum of energy ω as a particle. In other words, we can rephrase the state as: n_i identical particles occupying "mode i", each contributing energy ω. By mode we mean a single-particle state, which could be, for example, a momentum, position, or any other type of eigenstate of $\mathcal{H}$. (It may be helpful to relabel the state using greek letters $|n_\alpha\rangle$, where α labels the single-particle modes; as before α may any quantum number.)

What of the ground state $|0\rangle = |0, 0, \dots, 0\rangle$? In the oscillator toy model, this state has energy $E_0 = N\omega/2$, but in the second quantization language we interpret it as the vacuum state. It represents the Fock state where all possible quantum states are unoccupied.

The uncertainty principle forbids a state with simultaneously $\hat{x} = 0$ and $\hat{p} = 0$, so the oscillator can never sit quietly at the bottom of its well. To see this quantitatively, take the expectation value of a single-oscillator Hamiltonian in an arbitrary state $|\psi\rangle$ and denote $\Delta x^2 = \langle\psi| \hat{x}^2 |\psi\rangle - \langle\psi| \hat{x} |\psi\rangle^2$ and similarly for Δp:

$$\langle\psi| \hat{H} |\psi\rangle \;=\; \frac{\langle\psi| \hat{p}^2 |\psi\rangle}{2m} + \frac{m\omega^2 \langle\psi| \hat{x}^2 |\psi\rangle}{2} \;\geq\; \frac{(\Delta p)^2}{2m} + \frac{m\omega^2 (\Delta x)^2}{2}, \tag{8}$$

where we used $\langle\psi| \hat{x}^2 |\psi\rangle \geq (\Delta x)^2$ and likewise for $\hat{p}$. Using $(A+B)/2 \geq \sqrt{AB}$ we get :

$$\langle\psi| \hat{H} |\psi\rangle \geq \omega \Delta p \, \Delta x \geq \frac{\omega}{2}, \tag{9}$$

by virtue of the Heisenberg uncertainty relation $\Delta x \, \Delta p \geq 1/2$. So $\langle\psi| \hat{H} |\psi\rangle \geq \omega/2$ for anu state ψ. The oscillator ground state saturates this bound.

The uncertainty principle in second quantization is implicit in the the commutation relations, $[a, a^\dagger] = 1$. To see this, invert the definitions of a and $a^\dagger$:

$$\hat{x} \;=\; \frac{1}{\sqrt{2m\omega}}\,(a + a^\dagger), \qquad \hat{p} \;=\; i\sqrt{\frac{m\omega}{2}}\,(a^\dagger - a), \tag{10}$$

and compute

$$[\hat{x},\,\hat{p}] \;=\; \frac{i}{2}\,[a + a^\dagger,\; a^\dagger - a] \;=\; \frac{i}{2}([a, a^\dagger] - [a^\dagger, a]) \;=\; \frac{i}{2}\,(1 - (-1)) \;=\; i. \tag{11}$$

So $[\hat{x}, \hat{p}] = i$. The canonical commutator behind $\Delta x\,\Delta p \geq 1/2$ is contained in the creation and annihilation operators on the Fock space.

Symmetrization. This toy model has a very desirable property: $a_i^\dagger$ and a_i automatically create states that are either symmetric or antisymmetric under particle exchange. If we use the commutation relation $[a_i^\dagger, a_j^\dagger] = 0$ inherited from the oscillator analogy, we get

$$\left|0, \dots, n_i, \dots, n_j, \dots, 0\right\rangle = (a_i^\dagger)^{n_i}(a_j^\dagger)^{n_i}\,|0\rangle = (a_j^\dagger)^{n_j}(a_i^\dagger)^{n_i}\,|0\rangle = \left|0, \dots, n_j, \dots, n_i, \dots, 0\right\rangle . \tag{12}$$

Therefore, the state is automatically symmetric (bosonic) under particle exchange (the 2-particle case we had been working with is just the special case $n_i = n_j = 1$). The occupation numbers are also unbounded, $n_i \in \{0, 1, 2, \dots\}$, so arbitrarily many bosons may pile into a single mode.

For fermions we want the two-particle state to flip sign under exchange. To achieve, we would like our creation and annihilation operators to satisfy anticommutation relations instead:

$$\{a_i, a_j^\dagger\} = \delta_{ij}, \qquad \{a_i, a_j\} = \{a_i^\dagger, a_j^\dagger\} = 0. \tag{13}$$

With that, we get

$$\left|0, \dots, n_i, \dots, n_j, \dots, 0\right\rangle = (a_i^\dagger)^{n_i}(a_j^\dagger)^{n_j}\,|0\rangle = -\,(a_j^\dagger)^{n_j}(a_i^\dagger)^{n_i}\,|0\rangle = -\left|0, \dots, n_j, \dots, n_i, \dots, 0\right\rangle , \tag{14}$$

which is just the antisymmetrization property we wanted. Note how the anticommutation relations also imply that $a_i^\dagger a_i^\dagger = -a_i^\dagger a_i^\dagger$, so $a_i^\dagger a_i^\dagger = 0$. This is just the Pauli exclusion principle (built into the algebra of our creation and annihilation operators).

From now on, we will use the state $|n_i\rangle$ to denote a state with n_i particles in mode i: $|n_i\rangle = |0, \dots, n_i, \dots, 0\rangle$. If more than one mode is occupied, we will write $\left|n_i, n_j, \dots\right\rangle$, omitting the zeros for brevity.

With the language above, it is even easier to show that bosons like to pile up in the same state. Consider the transition amplitude for our system to go from the vacuum $|0\rangle$ to the state $|n_i\rangle$, which has n_i bosons in mode i. The amplitude must contain a factor of $(a_i^\dagger)^{n_i}$ for it to be non-zero, therefore

$$t_{0\to n_i} = \langle n_i|\,(a_i^\dagger)^{n_i}\,|0\rangle = \langle n_i|\,\sqrt{1}\sqrt{2}\dots\sqrt{n_i - 1}\sqrt{n_i}\,|n_i\rangle \implies \Gamma_{0\to n_i} \propto |t_{0\to n_i}|^2 \propto n_i!, \tag{15}$$

showing a factorial enhancement in the transition rate. Had we populated n_i different modes instead, no such factorial shows up. For fermions, only the latter possibility exists.

Fock Space. Formally, the space for identical-particle quantum mechanics with an arbitrary number of particles is called the Fock space,

$$\mathcal{F} \;=\; \bigoplus_{n=0}^{\infty} \mathcal{H}^{\otimes n}. \tag{16}$$

It represents the direct sum of all n-particle tensor products of the single-particle Hilbert space $\mathcal{H}$.[1] The $n = 0$ sector is one-dimensional and contains the vacuum $|0\rangle$; the $n = 1$ sector is $\mathcal{H}$ itself; the $n = 2$ sector is $\mathcal{H} \otimes \mathcal{H}$, and so on. Processes such as emissions, absorptions, decays change number of particles, so they move between sectors (hence the need for the direct sum in Eq. (16)).

We then equip $\mathcal{F}$ with operators that generalize the oscillator ladder operators $a, a^\dagger$ from a single mode to a whole family of single-particle modes. Let $\{|\alpha\rangle\}$ be an orthonormal basis of $\mathcal{H}$ (think of α as a momentum, position, or any quantum number labelling a single-particle state). For each α we introduce a creation operator $a^\dagger_\alpha$ and an annihilation operator a_α, with the vacuum defined by $a_\alpha |0\rangle = 0$ for every α. Successive creation then gives the Fock basis,

$$\left|n_{\alpha_1}, n_{\alpha_2}, \ldots\right\rangle \;=\; \frac{(a^\dagger_{\alpha_1})^{n_{\alpha_1}} (a^\dagger_{\alpha_2})^{n_{\alpha_2}} \cdots}{\sqrt{n_{\alpha_1}!\, n_{\alpha_2}! \cdots}} \, |0\rangle \,, \tag{17}$$

with n_{α_i} the occupation number of mode α_i. What remains is to fix the algebra of the $a_\alpha, a^\dagger_\alpha$. As we will see, choosing between a commutator and an anticommutator amounts to choosing between bosons and fermions.

Continuum Limit. The oscillator toy model is actually a simpler, discretized version of a (nonrelativistic) field theory. Our modes i are discrete, but if we want our modes to correspond to real observables like momentum or position, we need to take the continuum limit. For example, promoting i to a continuous variable such as momenutm $\mathbf{p}$, we get a continuous family of operators $a^\dagger_{\mathbf{p}}$ and $a_{\mathbf{p}}$ that satisfy commutation or anticommutation relations involving delta functions instead of Kronecker deltas:

$$[a_{\mathbf{p}}, a^\dagger_{\mathbf{p}'}] = (2\pi)^3 \delta(\mathbf{p} - \mathbf{p}'), \quad [a_{\mathbf{p}}, a_{\mathbf{p}'}] = [a^\dagger_{\mathbf{p}}, a^\dagger_{\mathbf{p}'}] = 0, \tag{18}$$

for bosons, and

$$\{a_{\mathbf{p}}, a^\dagger_{\mathbf{p}'}\} = (2\pi)^3 \delta(\mathbf{p} - \mathbf{p}'), \quad \{a_{\mathbf{p}}, a_{\mathbf{p}'}\} = \{a^\dagger_{\mathbf{p}}, a^\dagger_{\mathbf{p}'}\} = 0, \tag{19}$$

for fermions. Our full space of states is now spanned by states of the form $\left|n_{\mathbf{p}_1}, n_{\mathbf{p}_2}, \ldots\right\rangle$, where $n_{\mathbf{p}_i}$ is the number of particles in the momentum mode $\mathbf{p}_i$. The vacuum state $|0\rangle$ is the state with no particles, and it satisfies $a_{\mathbf{p}} |0\rangle = 0$ for all $\mathbf{p}$. The Fock space is spanned by the states obtained by applying creation operators to the vacuum state:

$$\left|n_{\mathbf{p}_1}, n_{\mathbf{p}_2}, \ldots\right\rangle = \frac{(a^\dagger_{\mathbf{p}_1})^{n_{\mathbf{p}_1}} (a^\dagger_{\mathbf{p}_2})^{n_{\mathbf{p}_2}} \cdots}{\sqrt{n_{\mathbf{p}_1}! n_{\mathbf{p}_2}! \cdots}} \, |0\rangle \,, \tag{20}$$

where $n_{\mathbf{p}_i}$ is the number of particles in the momentum mode $\mathbf{p}_i$. At this stage, we can make our first contact with "quantum" fields: if we want to create a particle at a given position $\mathbf{x}$, we can define the field operator $\hat{\psi}(\mathbf{x})$ as the Fourier transform of the creation and annihilation operators in momentum space[2]:

$$\hat{\psi}(\mathbf{x}) = \int \frac{d^3p}{(2\pi)^3} e^{i\mathbf{p}\cdot\mathbf{x}} a_{\mathbf{p}}, \quad \hat{\psi}^\dagger(\mathbf{x}) = \int \frac{d^3p}{(2\pi)^3} e^{-i\mathbf{p}\cdot\mathbf{x}} a^\dagger_{\mathbf{p}}. \tag{21}$$

The field operator $\hat{\psi}(\mathbf{x})$ annihilates a particle at position $\mathbf{x}$, while $\hat{\psi}^\dagger(\mathbf{x})$ creates a particle at position $\mathbf{x}$.[3] The commutation or anticommutation relations of the field operators can be

[1] The basis is built from product states, but physical states can still be superpositions that show entanglement between particles or even between sectors of different n.

[2] We could have called these $a_{\mathbf{x}}$ and $a^\dagger_{\mathbf{x}}$ instead, but the name field operator is more common.

[3] The factor of $(2\pi)^3$ in the Fourier transform is a matter of convention. It shows up here because we are using the normalization $\langle \mathbf{p}|\mathbf{p}'\rangle = (2\pi)^3 \delta(\mathbf{p} - \mathbf{p}')$.

derived from those of the creation and annihilation operators in momentum space, and they take the form

$$[\hat{\psi}(\mathbf{x}), \hat{\psi}^\dagger(\mathbf{y})] = \delta(\mathbf{x}-\mathbf{y}), \quad [\hat{\psi}(\mathbf{x}), \hat{\psi}(\mathbf{y})] = [\hat{\psi}^\dagger(\mathbf{x}), \hat{\psi}^\dagger(\mathbf{y})] = 0, \tag{22}$$

for bosons, and

$$\{\hat{\psi}(\mathbf{x}), \hat{\psi}^\dagger(\mathbf{y})\} = \delta(\mathbf{x}-\mathbf{y}), \quad \{\hat{\psi}(\mathbf{x}), \hat{\psi}(\mathbf{y})\} = \{\hat{\psi}^\dagger(\mathbf{x}), \hat{\psi}^\dagger(\mathbf{y})\} = 0, \tag{23}$$

for fermions.

The field $\hat{\psi}(\mathbf{x})$ is related to the single-particle wavefunction $\psi(\mathbf{x})$ in first quantization by the relation

$$\psi(\mathbf{x}) = \langle 0| \hat{\psi}(\mathbf{x}) |\psi\rangle_{\mathcal{F}}\,, \tag{24}$$

where we include a $\mathcal{F}$ in $|\psi\rangle_{\mathcal{F}}$ to emphasize that it is a state in the Fock space. It corresponds to a single particle occupying the state $|\psi\rangle$. In analogy to our Toy Model, we may as well have called it $|1_\psi\rangle = a^\dagger_\psi |0\rangle$, where $a^\dagger_\psi$ is the creation operator for the single-particle state $|\psi\rangle$.

To understand Eq. (24) equation, we can just interpret $\langle 0| \hat{\psi}(\mathbf{x})$ as $\langle \mathbf{x}|$. To be precise: within the one-particle sector (a single factor of $\mathcal{H}$ in our Fock space), $\hat{\psi}^\dagger(\mathbf{x})|0\rangle$ plays the role of the position eigenket $|\mathbf{x}\rangle$, and $\langle 0| \hat{\psi}(\mathbf{x})$ plays the role of $\langle \mathbf{x}|$. Then, $\psi(\mathbf{x}) = \langle \mathbf{x}|\psi\rangle_{\mathcal{F}}$ is just the expansion of the state $|\psi\rangle_{\mathcal{F}}$ in the position basis, which is exactly what we had in first quantization. Being more precise now,

$$|\psi\rangle_{\mathcal{F}} = \int d^3\mathbf{x}\, |\mathbf{x}\rangle \langle \mathbf{x}|\psi\rangle_{\mathcal{F}} \to \int d^3\mathbf{x}\, \psi(\mathbf{x})\, |\mathbf{x}\rangle = \int d^3\mathbf{x}\, \psi(\mathbf{x}) \hat{\psi}^\dagger(\mathbf{x})\, |0\rangle\,. \tag{25}$$

Note the absence of a hat on $\psi(\mathbf{x})$: the wavefunction is just a coefficient in the expansion of the state $|\psi\rangle_{\mathcal{F}}$ in the position basis. So, all the physics that we had encoded in the wavefunction in first quantization is still there.

Operators. So far so good, but we have not yet said anything about how to construct a theory other than one of uncoupled harmonic oscillators. We would also like to add dynamics to our system that allows transitions between different states in our Fock space (interactions between different particles, for example).

In the oscillator toy model, to build the Hamiltonian we could have easily guessed that $\hat{H} = \omega \sum_i (N_i + 1/2)$: it counts the number of particles in each mode and assigns an energy ω to each. The zero point energy is a constant then to be determined in some other way (in the oscillator case, it was calculated explicitly by solving Schrödinger's equation). The procedure of finding new operators and Hamiltonians in second quantization can be then generalized by analogy with the operators in the single-particle spaces. For example, take some operator K that acts on the single-particle Hilbert space $\mathcal{H}$. We can ask what is the corresponding operator $\mathcal{K}$ that acts on the Fock space $\mathcal{F}$ and captures the same physics. To find $\mathcal{K}$, we write

$$\mathcal{K} = \sum_{\alpha,\beta} \langle \alpha| K |\beta\rangle\, a^\dagger_\alpha a_\beta = \sum_{\alpha,\beta} K_{\alpha\beta} a^\dagger_\alpha a_\beta, \tag{26}$$

where $K_{\alpha\beta} = \langle \alpha| K |\beta\rangle$ is the matrix element of K in the α basis and $a^{(\dagger)}_\alpha$ is the creation/annihilation operator in that same basis. For a continuous basis,

$$\mathcal{K} = \int d\alpha\, d\beta\, \langle \alpha| K |\beta\rangle\, a^\dagger_\alpha a_\beta. \tag{27}$$

This is a good start, but we may not know the annihilation and creation operators in the α basis. So a more useful form comes from writing these in terms of creation and annihilation operators in yet a different basis, such as the position basis, or any other basis β. To convert between bases, we can use

$$a_\alpha = \int d\beta \, \langle\alpha|\beta\rangle \, a_\beta, \quad a_\alpha^\dagger = \int d\beta \, \langle\beta|\alpha\rangle \, a_\beta^\dagger. \tag{28}$$

This is precisely the relation between the momentum-space and position-space creation and annihilation operators we have in Eq. (21).

With this, we can write any operator in, for example, the position basis:

$$\mathcal{K} = \int d\alpha \, d\beta \, \langle\alpha| K |\beta\rangle \, a_\alpha^\dagger a_\beta = \int d^3\mathbf{x} \, d^3\mathbf{y} \, d\alpha \, d\beta \, \langle\alpha| K |\beta\rangle \, \langle\mathbf{x}|\alpha\rangle \, \langle\beta|\mathbf{y}\rangle \, \hat{\psi}^\dagger(\mathbf{x})\hat{\psi}(\mathbf{y}) \tag{29}$$

$$= \int d^3\mathbf{x} \, d^3\mathbf{y} \, \langle\mathbf{x}| K |\mathbf{y}\rangle \, \hat{\psi}^\dagger(\mathbf{x})\hat{\psi}(\mathbf{y}), \tag{30}$$

where $\langle\mathbf{x}| K |\mathbf{y}\rangle$ is the kernel of the operator K in position space. For example, the kinetic energy operator has the kernel

$$\langle\mathbf{x}| K |\mathbf{y}\rangle = \langle\mathbf{x}| \frac{\hat{p}^2}{2m} |\mathbf{y}\rangle = -\frac{1}{2m}\nabla_\mathbf{x}^2\delta^3(\mathbf{x}-\mathbf{y}) = -\frac{1}{2m}\nabla_\mathbf{y}^2\delta^3(\mathbf{x}-\mathbf{y}), \tag{31}$$

where the two forms are equivalent as distributions inside the integral. Almost immediately we get the Hamiltonian of a free particle in second quantization:

$$\hat{H} = \int d^3x \, \hat{\psi}^\dagger(\mathbf{x}) \left(-\frac{\nabla^2}{2m}\right) \hat{\psi}(\mathbf{x}). \tag{32}$$

In momentum space, the Hamiltonian is diagonal and takes the form

$$\hat{H} = \int \frac{d^3p}{(2\pi)^3} \frac{\mathbf{p}^2}{2m} a_\mathbf{p}^\dagger a_\mathbf{p} = \int \frac{d^3p}{(2\pi)^3} \frac{\mathbf{p}^2}{2m} N_\mathbf{p}, \tag{33}$$

where $N_\mathbf{p} = a_\mathbf{p}^\dagger a_\mathbf{p}$ is the number operator for the momentum mode $\mathbf{p}$. We are just counting the number of particles of energy $\mathbf{p}^2/2m$ in each momentum mode and summing over all modes to get the total energy.

Note how our formalism now requires a basis. This is a key difference between first and second quantization: in first quantization, the state $|\psi\rangle$ is a vector in a Hilbert space, and the Hamiltonian $\hat{H}$ is an operator that acts on this vector. In second quantization, the state $|\psi\rangle$ is now a vector in the Fock space, and the Hamiltonian $\hat{H}$ is now expressed in terms of $\hat{\psi}(\mathbf{x})$ and $\hat{\psi}^\dagger(\mathbf{x})$ and also acts on the Fock space. Momentum and position are now "demoted" to labels of the field operators. Operations move our occupation numbers around, but they do not change the basis of the single-particle Hilbert space $\mathcal{H}$. One may still define position and momentum operators, such as

$$\hat{\mathbf{X}} = \int d^3x \, \mathbf{x}\hat{\psi}^\dagger(\mathbf{x})\hat{\psi}(\mathbf{x}), \quad \hat{\mathbf{P}} = \int d^3x \, \hat{\psi}^\dagger(\mathbf{x})(-i\nabla)\hat{\psi}(\mathbf{x}). \tag{34}$$

They act on the Fock space and can be used to compute expectation values and matrix elements, but they are not the building blocks of the theory.

Interactions and Dynamics. The general form of our Hamiltonians will be

$$\hat{H} = \sum_{\alpha,\beta} h_{\alpha\beta} a_\alpha^\dagger a_\beta + \frac{1}{2} \sum_{\alpha,\beta,\gamma,\delta} V_{\alpha\beta\gamma\delta} a_\alpha^\dagger a_\beta^\dagger a_\gamma a_\delta + \cdots, \tag{35}$$

where $h_{\alpha\beta} = \langle\alpha| \hat{h} |\beta\rangle$ is the single-particle Hamiltonian matrix element (for example, the kinetic term discussed above) while $V_{\alpha\beta\gamma\delta} = \langle\alpha\beta| \hat{V} |\gamma\delta\rangle$ is the two-particle interaction matrix element. The continuum case is obtained as above. The first term describes the free evolution of the particles, while the second term describes interactions between pairs of particles. Note how a two-body interaction term involves two creation and two annihilation operators, which corresponds to the process of two particles interacting and changing their states:

$$a^\dagger_\alpha a^\dagger_\beta a_\gamma a_\delta \left| n_\alpha, n_\beta, n_\gamma, n_\delta \right\rangle \to \left| n_\alpha + 1, n_\beta + 1, n_\gamma - 1, n_\delta - 1 \right\rangle , \tag{36}$$

where we have assumed that the initial state has at least one particle in modes γ and δ to be annihilated, and we have created one particle in modes α and β. Higher-order terms can describe three-body interactions, four-body interactions, etc, depending on the physical system we are trying to model.

> Show that the total particle number N is conserved in quantum mechanics by showing that the operator $\hat{N} = \sum_\alpha a^\dagger_\alpha a_\alpha$ commutes with the Hamiltonian $\hat{H}$, i.e., $[\hat{N}, \hat{H}] = 0$.